

ROHAN MISTRY

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SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, HTML/CSS, SQL, R
Frameworks & Libraries: React, Node.js, Next.js, FastAPI, TensorFlow, PyTorch, Pandas, NumPy, Scikit-Learn
Tools & Platforms: Git, Docker, AWS, PostgreSQL, MySQL, Kubernetes, Maven, Tomcat, Tableau

CERTIFICATIONS

AWS Certified Solutions Architect – Associate (SAA-C03) | Amazon Web Services Mar 2026

EDUCATION

University of California, Irvine Sep 2021 – Jun 2025
B.S. Computer Science – Specialization: Intelligent Systems GPA: 3.7

- Relevant Coursework:** Data Structures & Algorithms, Artificial Intelligence, Machine Learning & Data Mining, Database Management, Software Engineering, Operating Systems, Compilers, Information Retrieval

EXPERIENCE

Research Data Engineer | Humanity Unleashed Initiative Sep 2024 – Jan 2025

- Engineered automated **Python** data pipelines to extract and standardize 800,000+ economic time series from FRED API and government databases, creating unified JSONL datasets for **multimodal AI model training**.
- Collaborated with research teams on foundation model architectures and time series forecasting, conducting literature reviews and implementing scalable data infrastructure using cloud storage.

Software & Data Intern | MKS Instruments Oct 2023 – Sep 2024

- Developed and deployed automated **Python** web scraping pipelines to streamline market research processes, reducing manual data collection by 75% and accelerating competitive insights.
- Delivered high-quality datasets, managed **SAP** inventory analyses, and created marketing collateral for new product launches.

Robotics Operations Intern | Starship Technologies Sep 2022 – Jan 2023

- Operated and maintained 50+ autonomous delivery robots, performing system diagnostics, sensor calibration, and troubleshooting to ensure 95% uptime and resolve real-time hardware/software issues.

PROJECTS

NFL Mock Draft Simulator
React, FastAPI, Python, HTML/CSS, JavaScript, PostgreSQL, SQLAlchemy, Netlify, Render, Supabase

- Built full-stack web application simulating the NFL Draft with real-time team management and drafting logic.
- Automated CPU drafting using weighted algorithms for positional needs and player rankings.
- Integrated **React** frontend, **FastAPI** backend, and **PostgreSQL** in cloud architecture supporting RESTful APIs, CRUD operations, and reliable data synchronization via connection pooling.

Fabflix
Java, JavaScript, MySQL, jQuery, AJAX, Docker, AWS, Maven, Tomcat, JDBC, Kubernetes, JMeter

- Engineered cloud-native platform to browse/purchase from catalog of 15,000 films.
- Developed robust architecture with **Tomcat**, **Java** servlets, Maven, and **MySQL**, implementing full-text search with autocomplete, session-based checkout, and secure authentication with HTTPS, AJAX, and reCAPTCHA.
- Optimized performance with MySQL connection pooling, Apache load balancing, and JMeter performance testing.
- Deployed **Kubernetes** cluster on **AWS (EC2)** using **Docker** containers for horizontal scalability.

Traffic Signal Control Using Reinforcement Learning
Python, PyTorch, RLLib, Pandas, NumPy, JavaScript, CityFlow

- Developed AI-driven multi-agent RL framework using MAPPO algorithms to optimize traffic signals, reducing congestion and increasing lane throughput in **CityFlow** simulations.
- Designed custom state/action representations and dynamic reward functions.
- Leveraged **Ray RLLib** for training/tuning, achieving measurable gains in traffic flow metrics on real-world datasets.

StrataScratch Exploratory Data Analysis
Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, DeepNote, Alteryx Designer

- Performed end-to-end EDA pipeline on Dublin Airbnb market, cleaning and visualizing data with **Python** libraries to uncover actionable insights on supply, demand, and pricing trends.
- Trained decision tree models with 89% predictive accuracy using **Scikit-learn**.
- Awarded *Best Analysis* at UC Irvine Datathon 2024. Collaborated via DeepNote/Alteryx for analytics workflows.